

Fundamental of Circuit Analysis

频率特性与谐振现象

Huarui Yin

Department of Electronics and Information Science
University of Science and Technology of China
Hefei, Anhui, 230027

内容简介

认识电路特性和电源频率密切相关

- 网络函数和频率特性
- RLC 串联电路频率特性
- 串联谐振电路
- 并联谐振电路
- 作业：P193 2 3 4 5 6 9 11 12 13 19

网络函数和频率特性

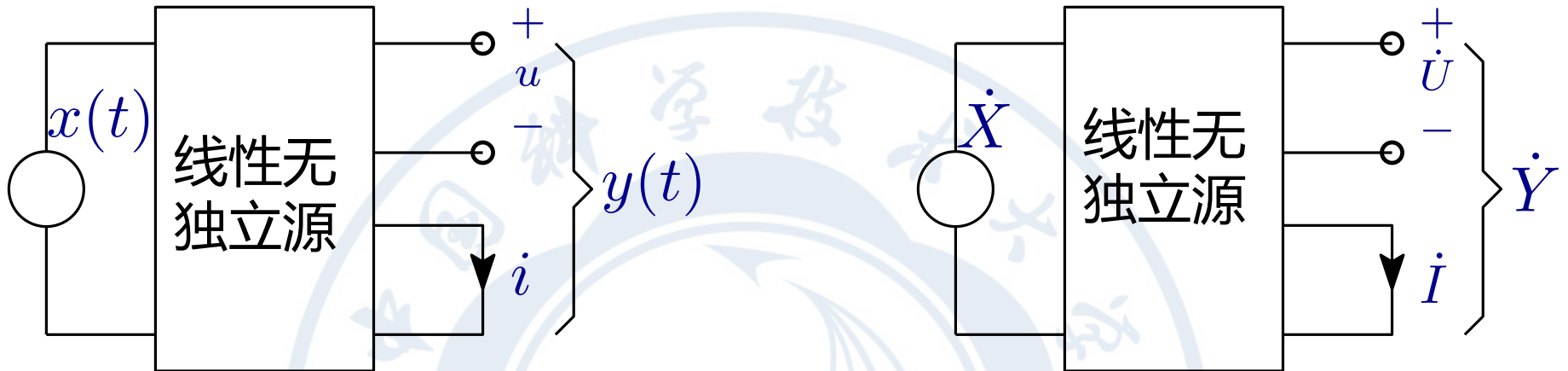
齐次定理：一个线性直流电路仅仅有一个电源，则任何支路的电压或者电流都于这个电源的源电压或者源电流成正比。

- 对于正弦电流电路, 所有的电路变量使用相量形式表达后, 上述定理依然成立。若定义源电压或者源电流为 $\dot{X}$, 输出为 $\dot{Y}$, 则有:

$$H(j\omega) = \frac{\dot{Y}}{\dot{X}}$$

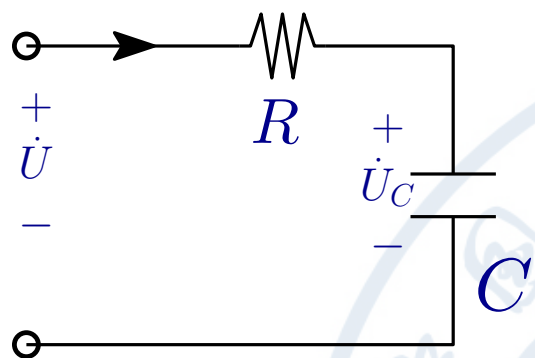
- $H(j\omega)$: 网络函数

网络函数



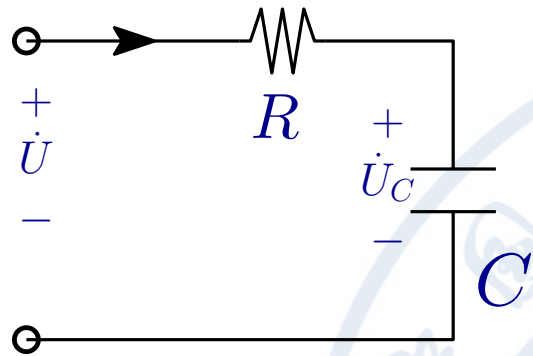
激励 $(x(t), \dot{X})$	响应 $(y(t), \dot{Y})$	转移函数
电流	电流	转移电流比
电流	电压	转移电阻
电压	电流	转移电导
电压	电压	转移电压比

网络函数-RC 电路



$$H(j\omega) = \frac{\dot{U}_C}{\dot{U}} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega CR}$$

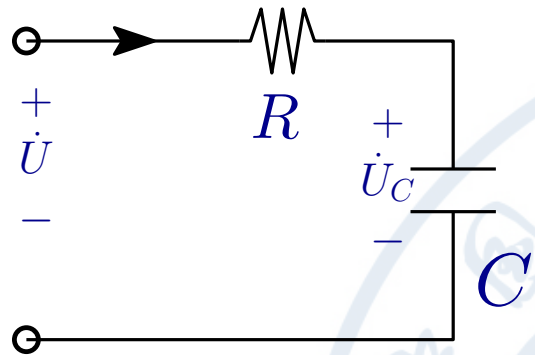
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■ 网络函数取决于**电路结构、元件参数以及电源频率**.

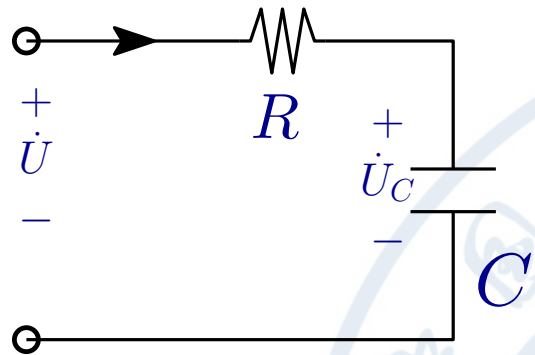
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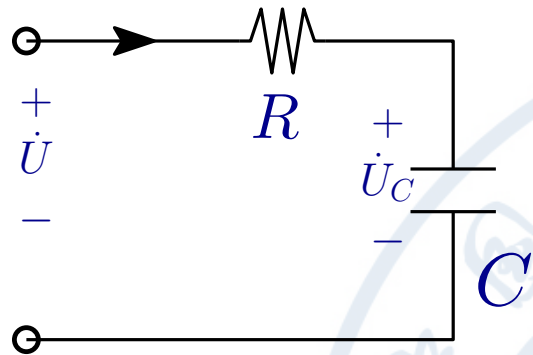
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网络函数-RC 电路

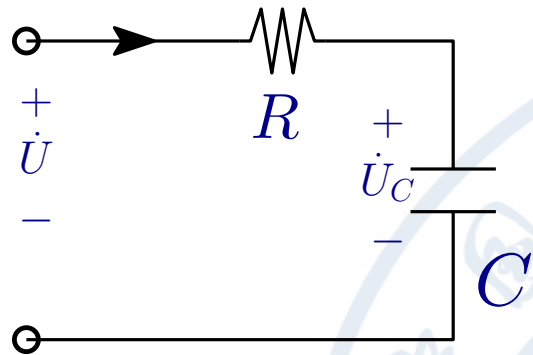


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幅度频率响应： $|H(j\omega)|$ ，相位频率响应： $\theta(\omega)$

网络函数-RC 电路

■ 固有频率，自然频率： $\omega_0 = \frac{1}{RC}$

$$H(j\omega) = \frac{1}{1+j\omega/\omega_0} = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_0}\right)^2}} \angle \left\{ -\arctan\left(\frac{\omega}{\omega_0}\right) \right\}$$

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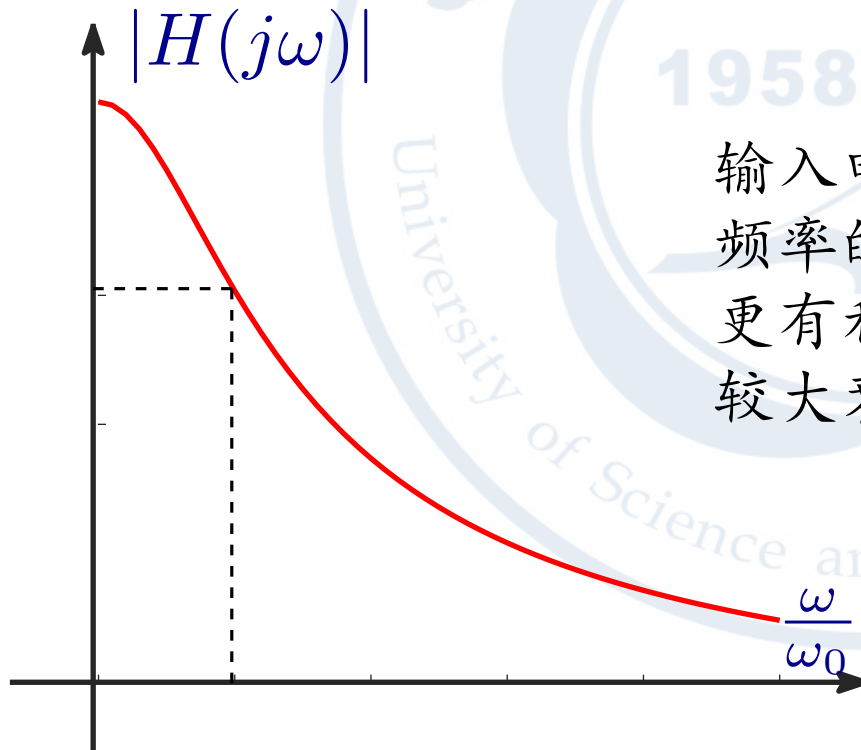
幅度频率响应

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输入电压有效值保持不变的情况，随着频率的升高，输出电压 U_C 减小。该电路更有利于低频信号通过，对高频信号有较大衰减，称为**低通网络**。

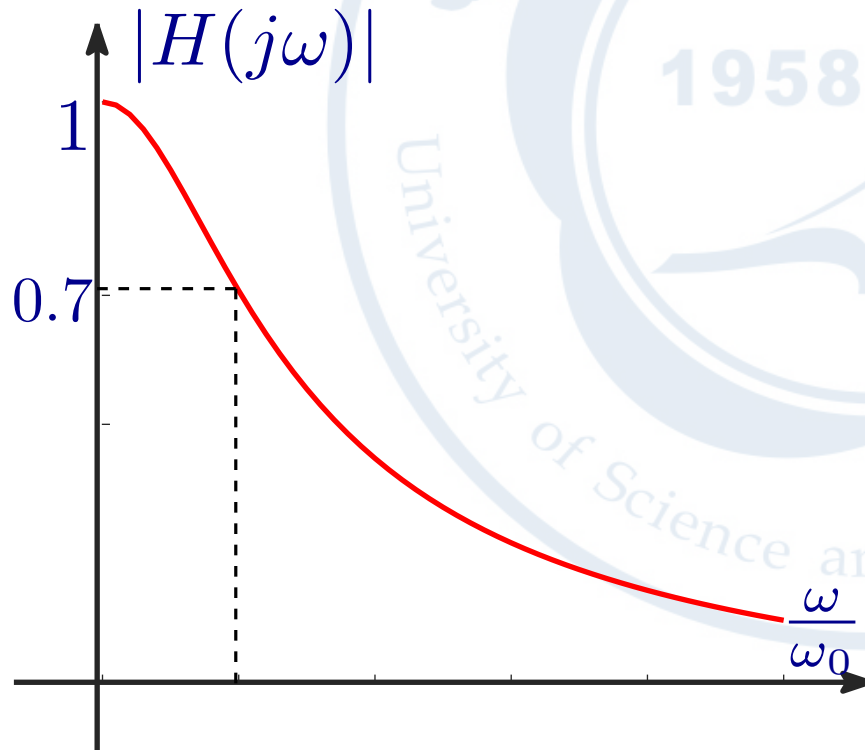
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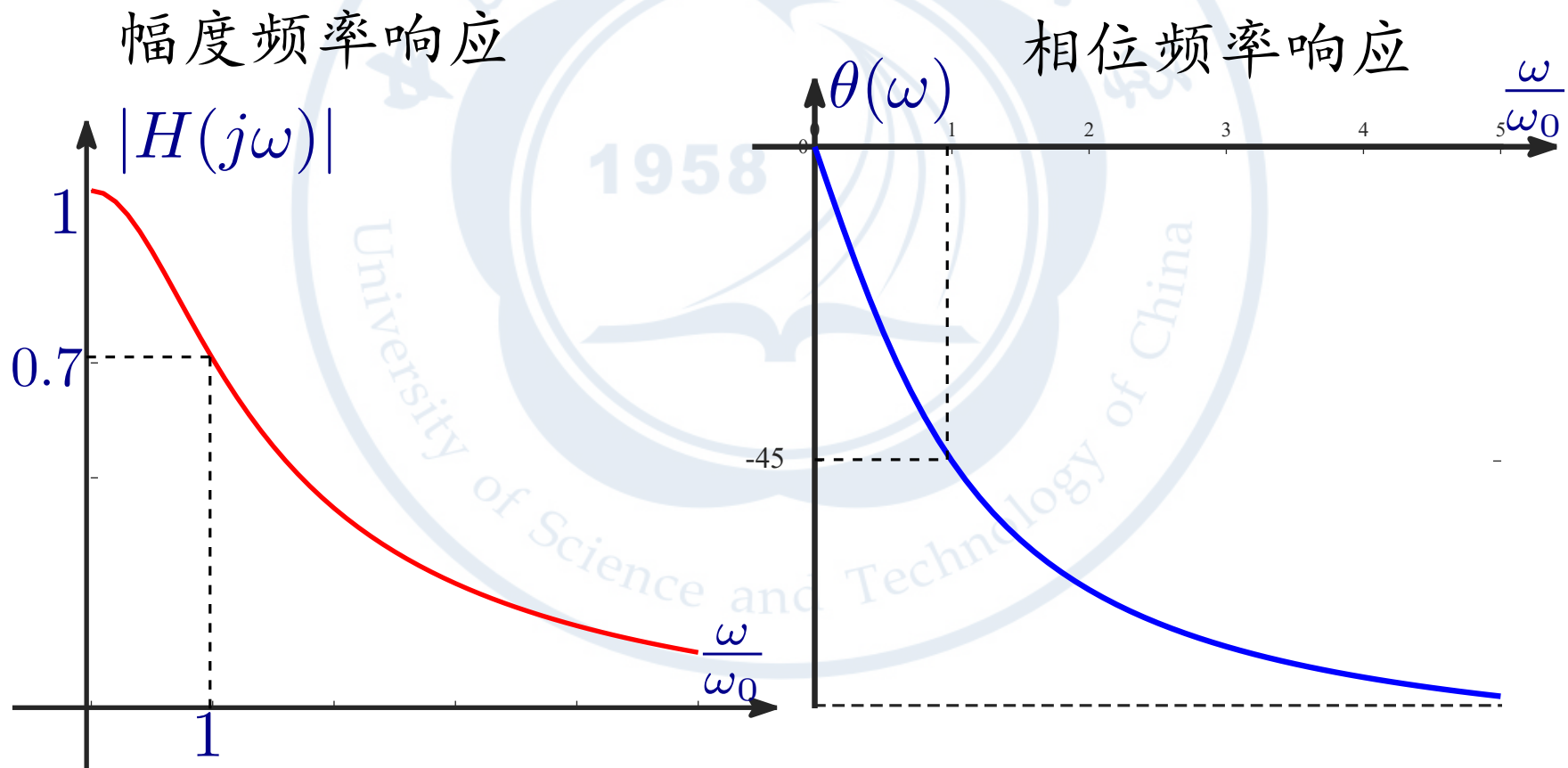
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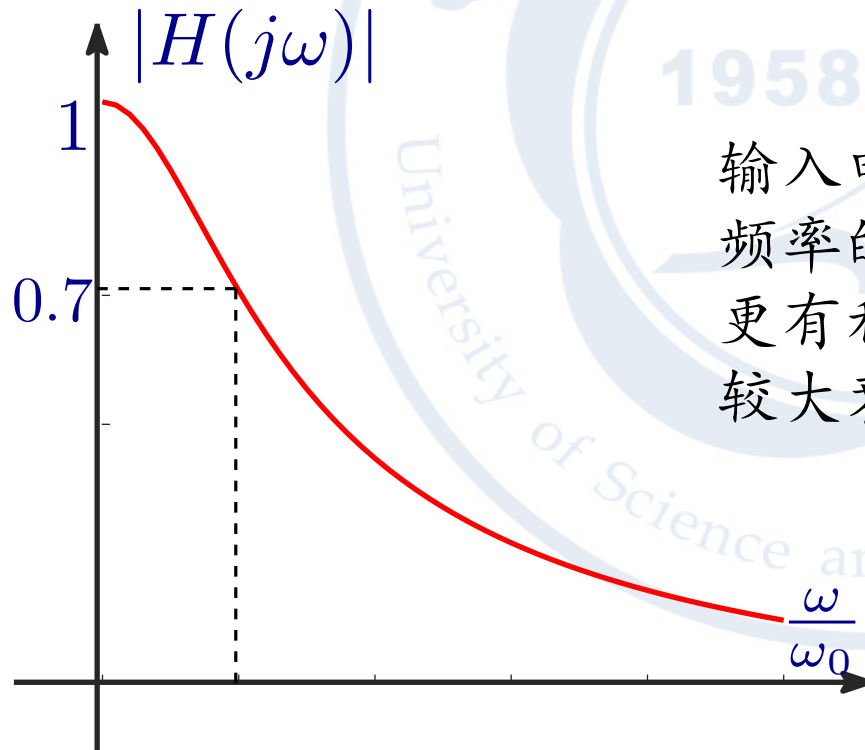
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幅度频率

幅频特性分类

全通网络	低通网络	高通网络	带通网络	带阻网络
—	LPF	HPF	BPF	BSF



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相关定义

网络函数的模下降到最大值的 $1/\sqrt{2}$ 所对应的频率	截止频率
所有的幅度频率响应比最大值的 $1/\sqrt{2}$ 大的频率点构成的点集	通带
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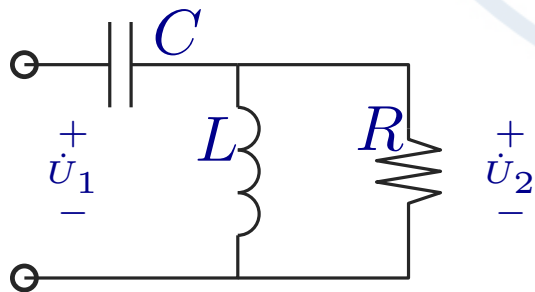
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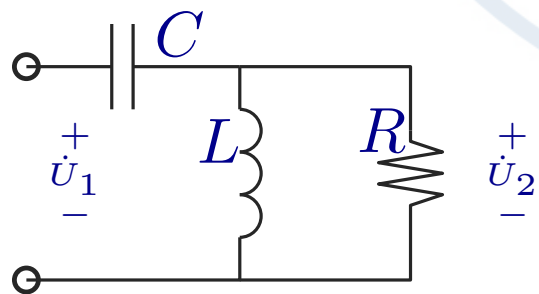
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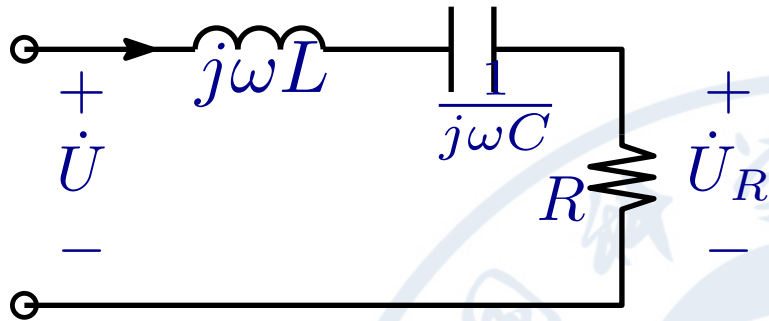
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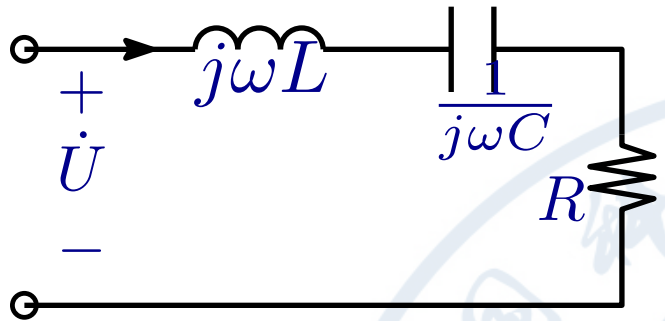


$$H(j\omega) = \frac{\dot{U}_2}{\dot{U}_1} = \frac{-\omega^2}{-\omega^2 + j\frac{\omega}{RC} + \frac{1}{LC}}$$

RLC 串联电路的频率特性

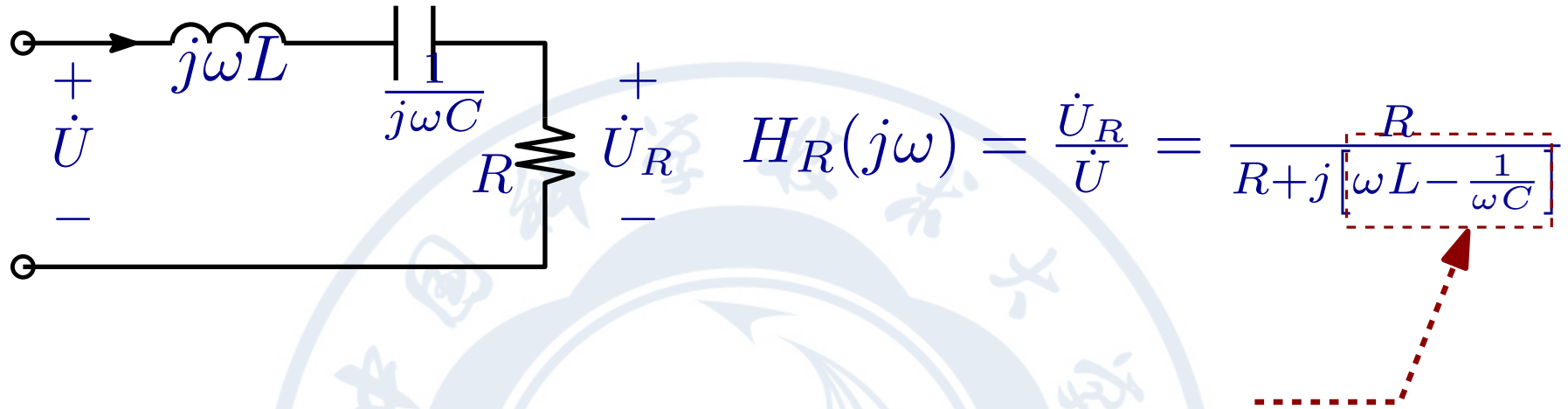


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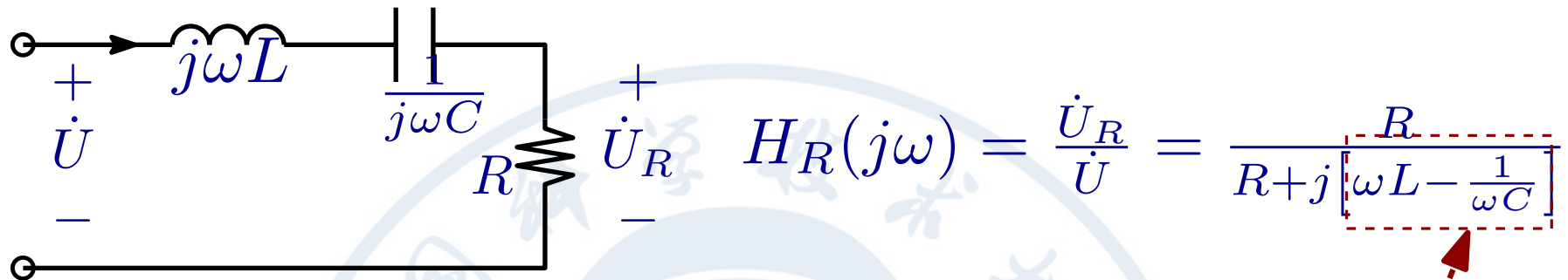


$$H_R(j\omega) = \frac{\dot{U}_R}{\dot{U}} = \frac{R}{R + j\left[\omega L - \frac{1}{\omega C}\right]}$$

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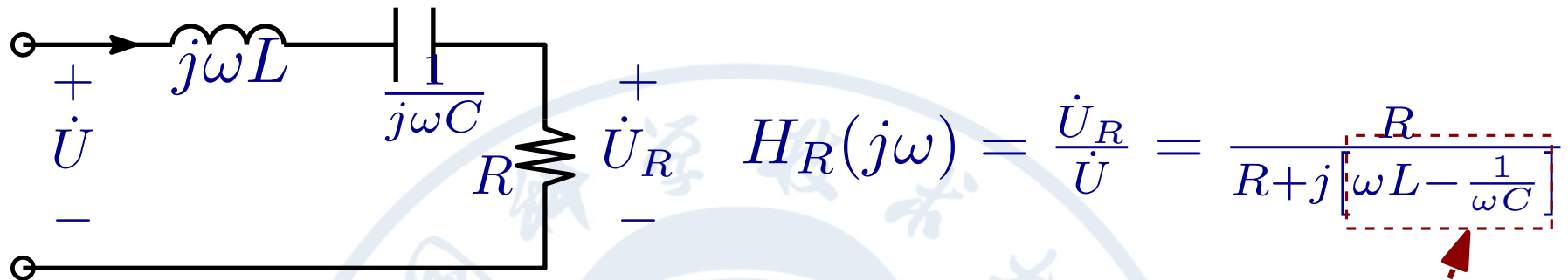


RLC 串联电路的频率特性



电容与电感作用相互抵消。抵消效果随着频率的不同而发生变化。

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$H_R(j\omega)$ 与频率相关的若干定义

谐振角频率	$\omega_0 = \frac{1}{\sqrt{LC}}$	电容与电感作用抵消
特性阻抗	$\rho = \omega_0 L = \frac{1}{\omega_0 C}$	谐振时感抗(容抗)模
品质因数	$Q = \frac{\rho}{R}$	谐振时感抗(容抗)电阻比

RLC 串联电路的频率特性

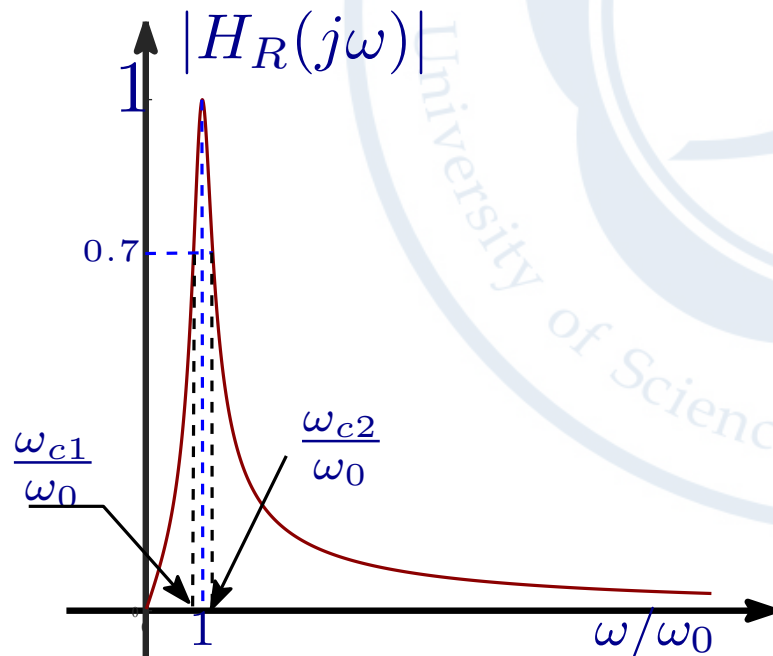
■ 幅度频率响应

$$|H_R(j\omega)| = \frac{1}{\sqrt{1 + \frac{1}{R^2} \left(\omega L - \frac{1}{\omega C} \right)^2}} = \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}}$$

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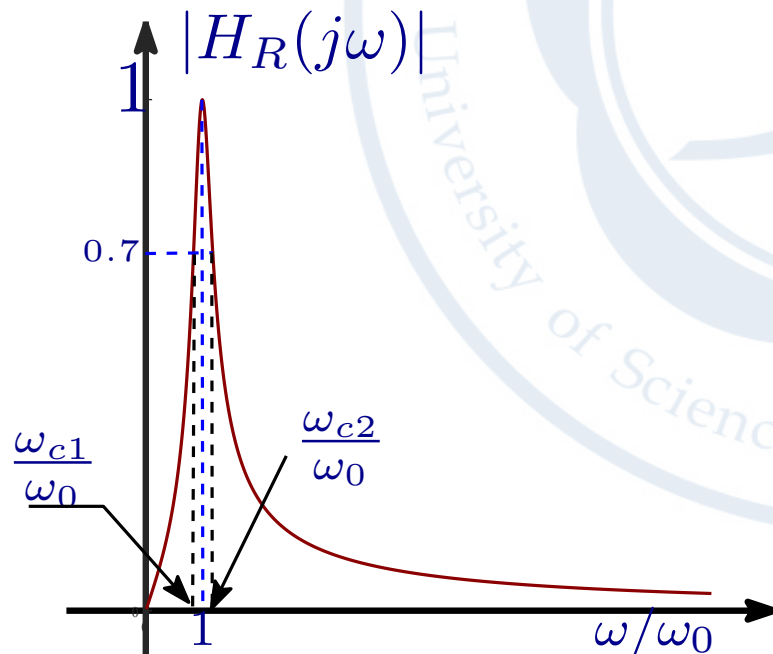
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1. $\omega/\omega_0 \rightarrow 1, |H_R(j\omega)| = 1 \rightarrow \max$
2. $\omega/\omega_0 \rightarrow \infty, |H_R(j\omega)| \rightarrow 0$
3. $\omega/\omega_0 \rightarrow 0, |H_R(j\omega)| \rightarrow 0$
4. $\omega \in [\omega_0, \infty), |H_R(j\omega)|$ 单调减
5. $\omega \in [0, \omega_0), |H_R(j\omega)|$ 单调增

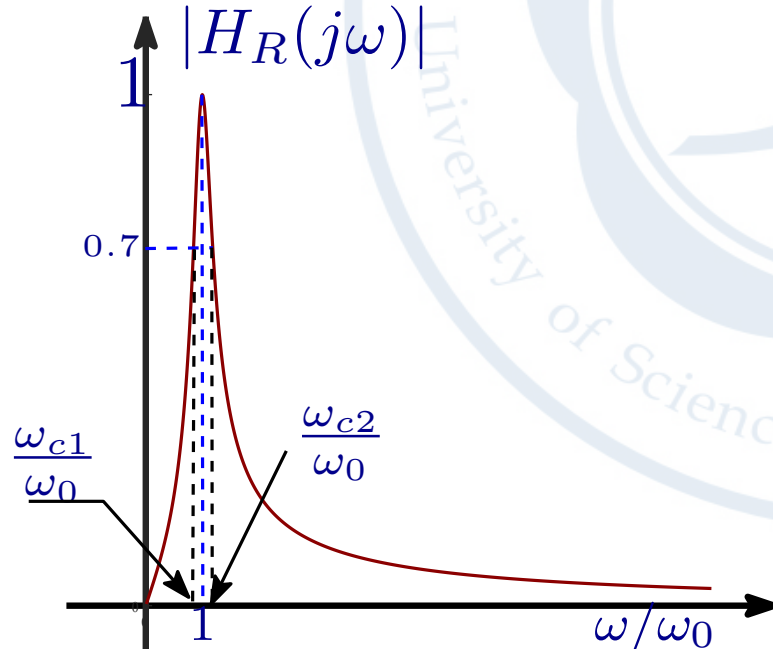
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■ 相位频率响应

$$\theta_R(\omega) = -\arctan \left\{ Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \right\}$$



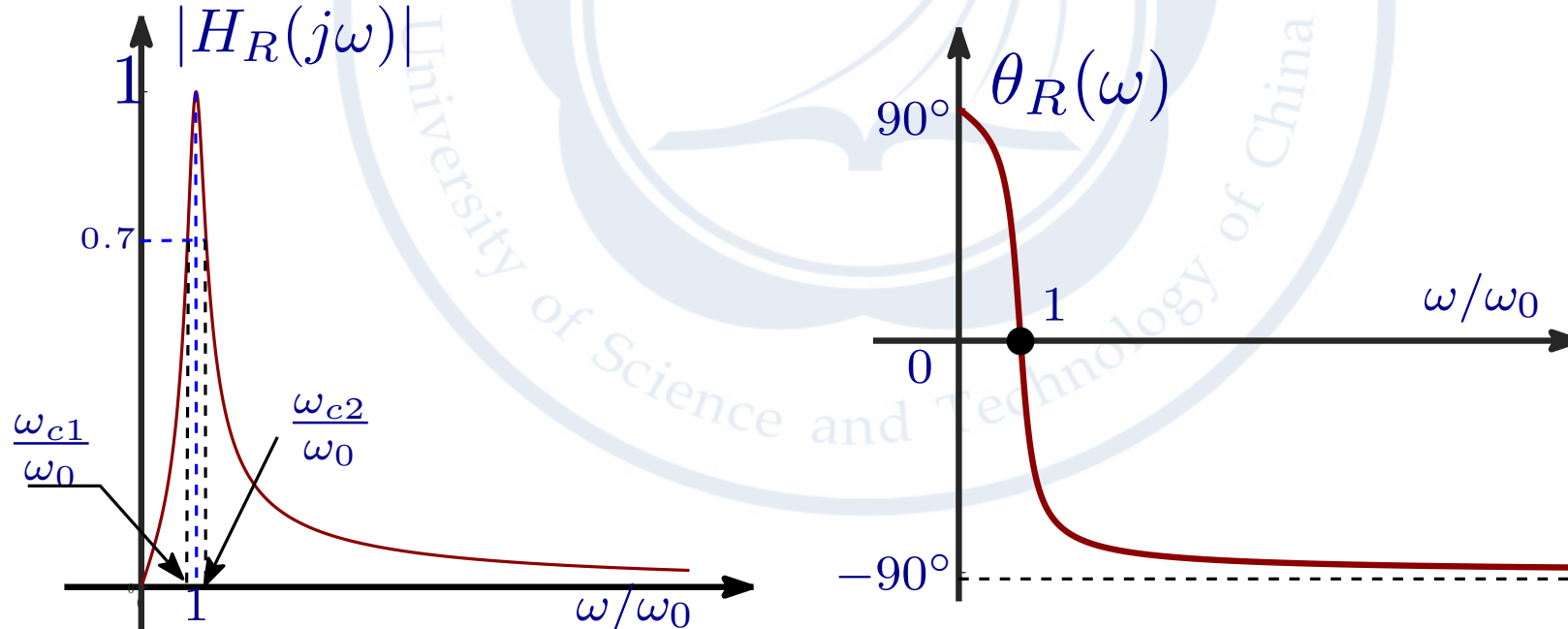
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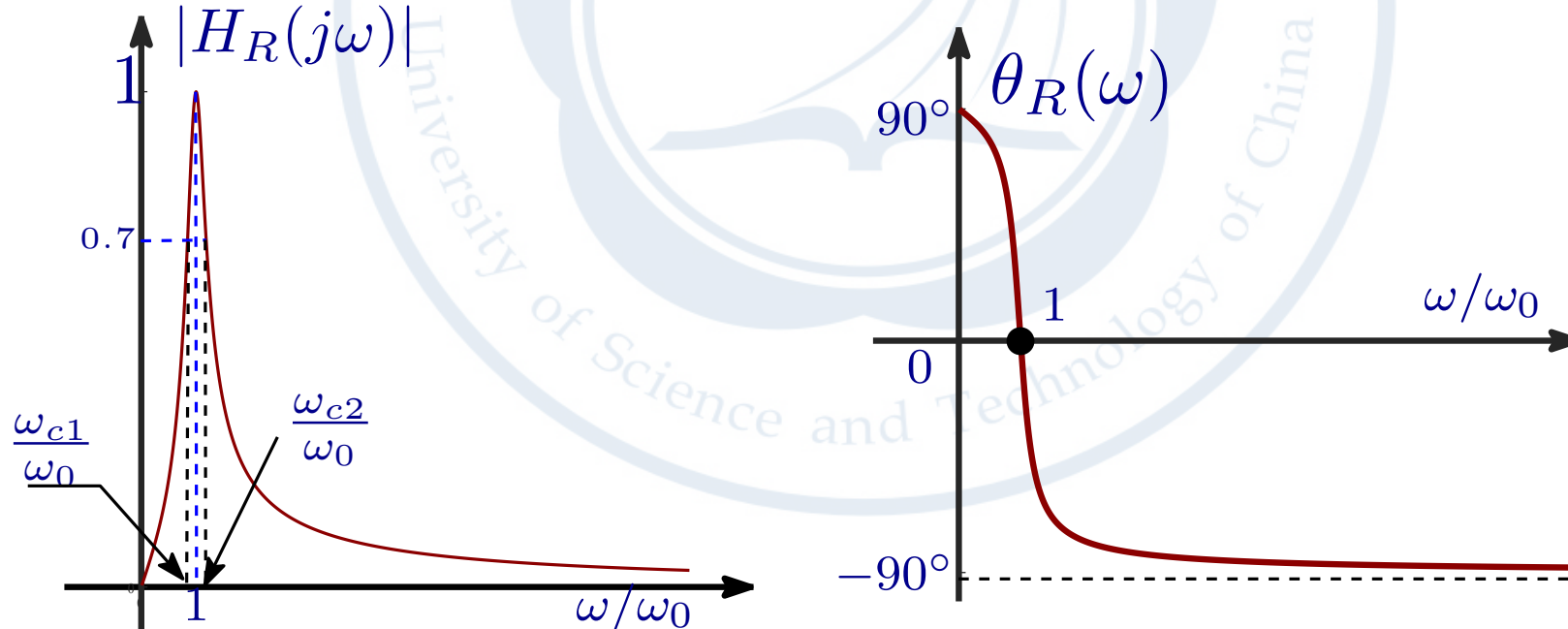
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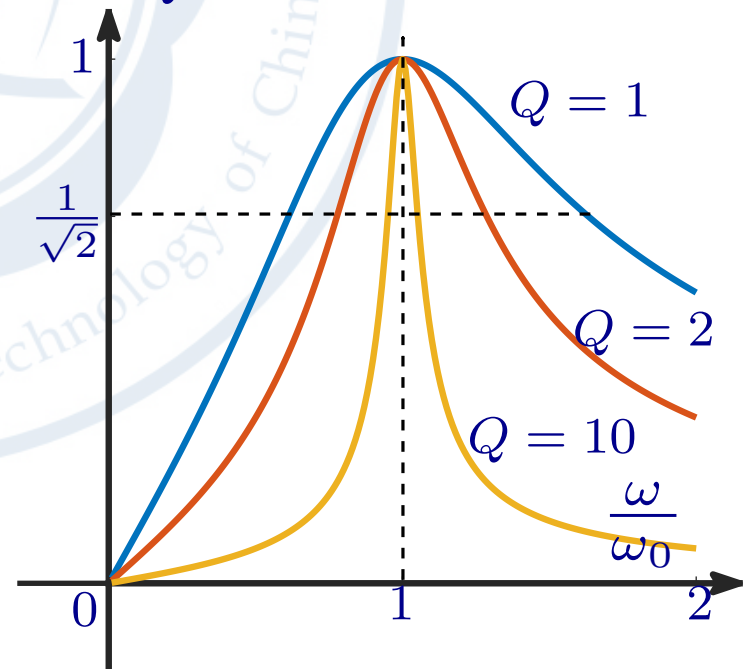
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$$\rightarrow Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) = \pm 1 \rightarrow |H_R(j\omega)| = 1/\sqrt{2}$$

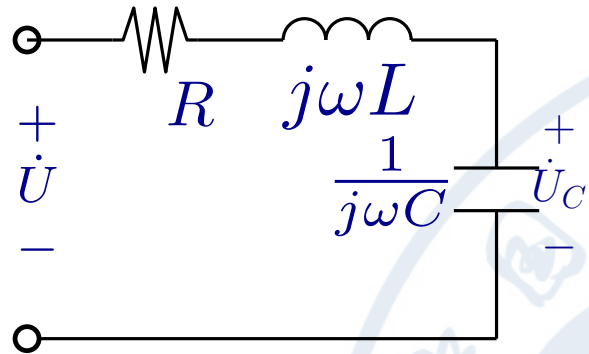
$$\rightarrow \omega_{c1}, \omega_{c2} = \omega_0 \left(\pm \frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$

$$\rightarrow \text{通带带宽: } \Delta\omega = \omega_{c1} - \omega_{c2} = \frac{\omega_0}{Q}$$

■ Q 对相对带宽有着直接的影响



RLC 串联电路的频率特性



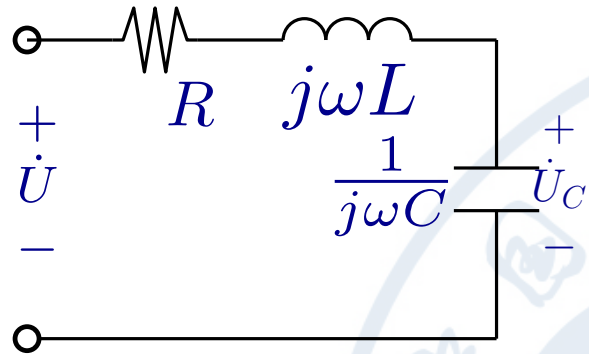
■ $H_C(j\omega)$ 幅度频率响应

$$|H_C(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega}{\omega_0}\right)^2}}$$

■ $H_C(j\omega)$ 相位频率响应

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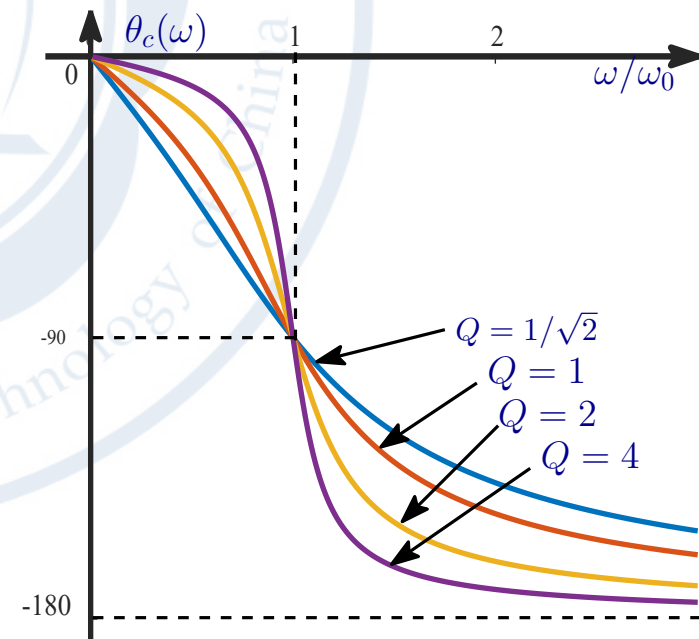
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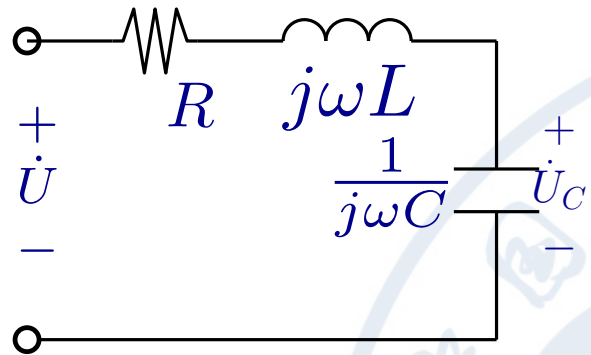
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ω/ω_0	$\theta_c(\omega)$
0	0°
1	-90°
∞	-180°



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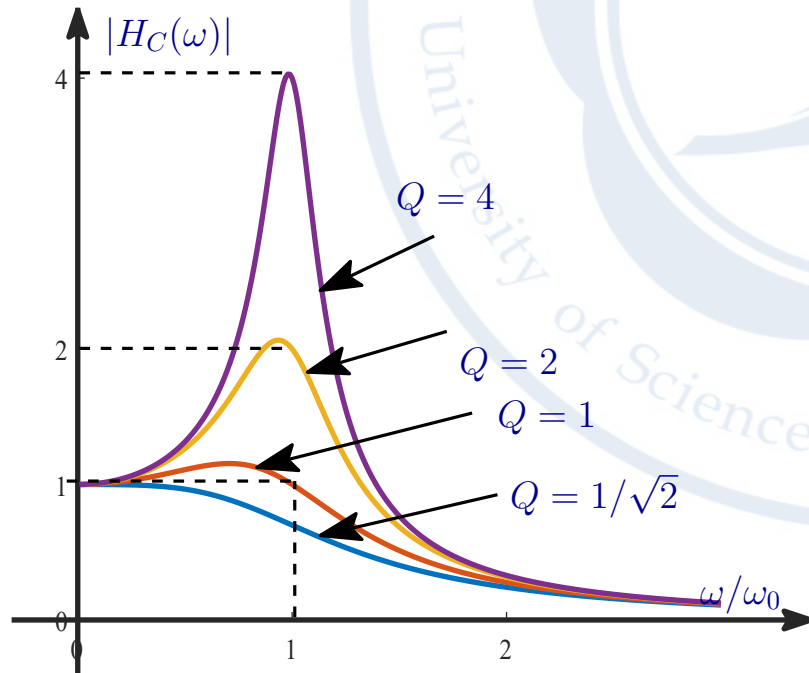


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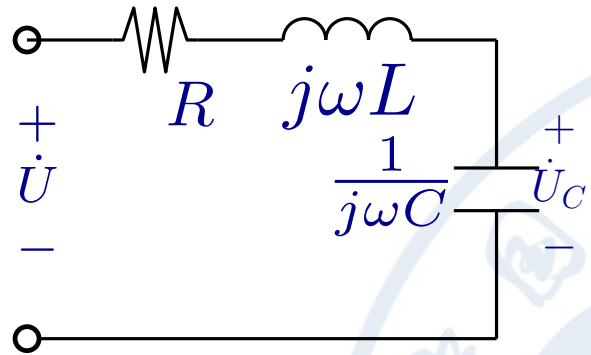


■ $Q > 1/\sqrt{2}$, 系统能够存在极大值

■ 极大值对应频率值略小于 ω_0 , 随着 $Q \rightarrow \infty$, 越来越接近于 ω_0

■ $Q > 1$ 时, 带通滤波器;
 $Q < 1$ 低通滤波器

RLC 串联电路的频率特性

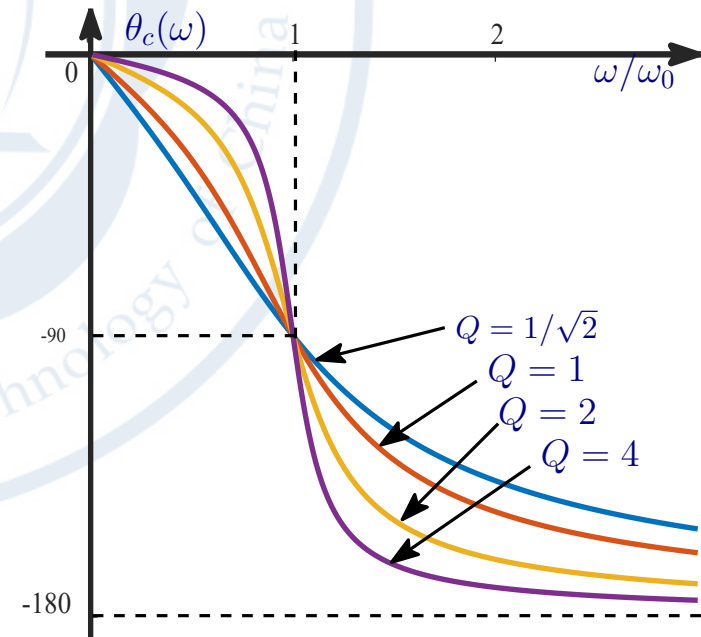
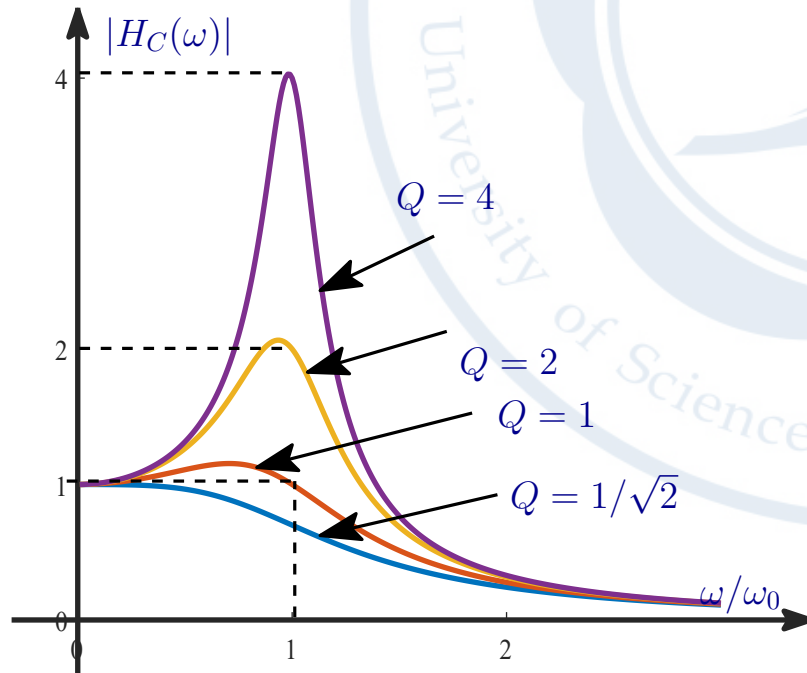


■ $H_C(j\omega)$ 幅度频率响应

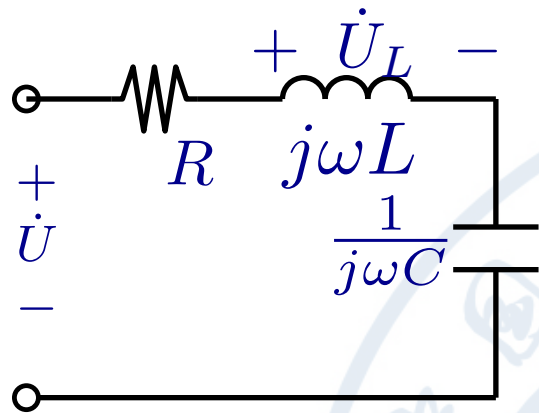
$$|H_C(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega}{\omega_0}\right)^2}}$$

■ $H_C(j\omega)$ 相位频率响应

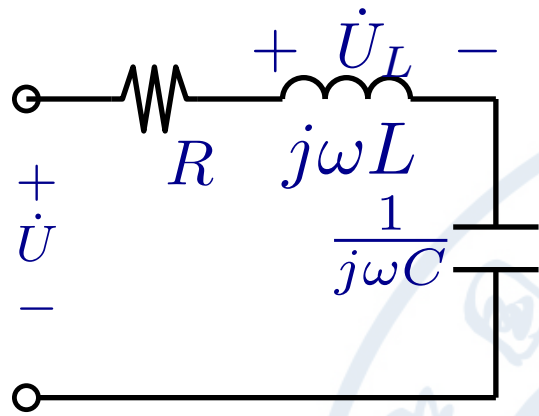
$$\theta_C(\omega) = -\arctan \frac{1}{Q \left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0} \right)}$$



RLC 串联电路的频率特性

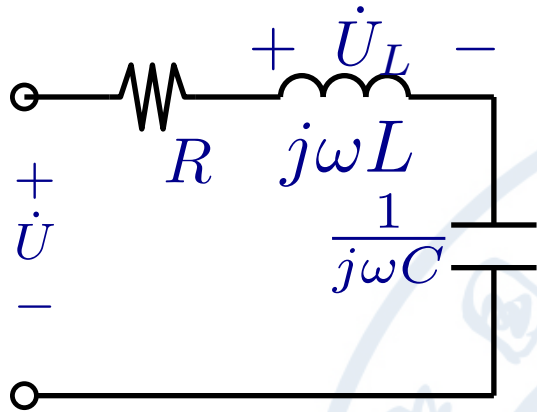


RLC 串联电路的频率特性



$$H_L(j\omega) = \frac{1}{\left[1 - \left(\frac{\omega_0}{\omega}\right)^2\right] - j\frac{1}{Q}\left(\frac{\omega_0}{\omega}\right)}$$

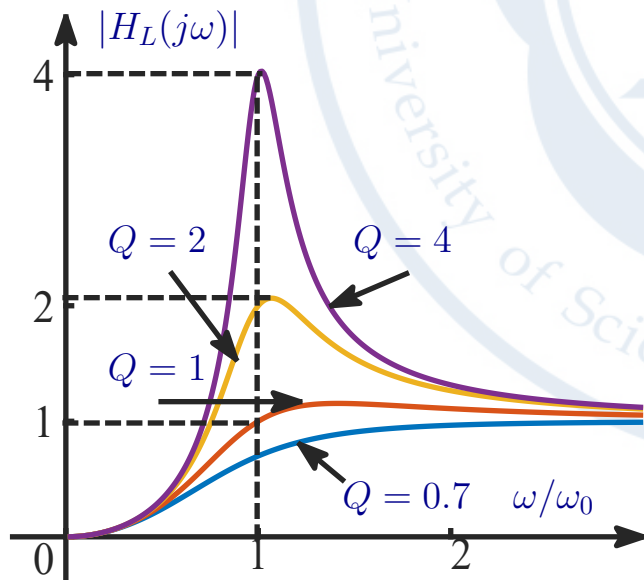
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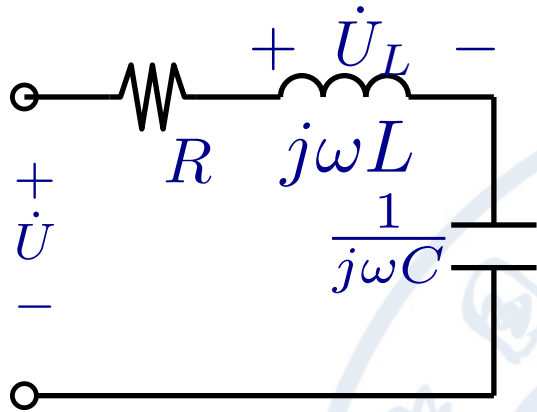
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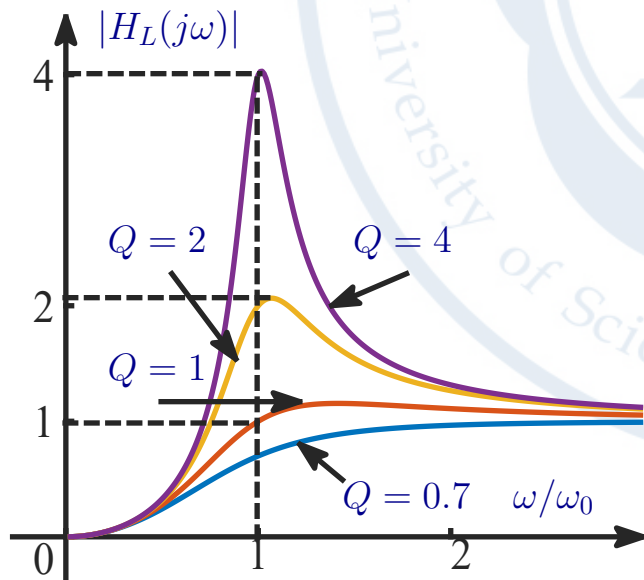
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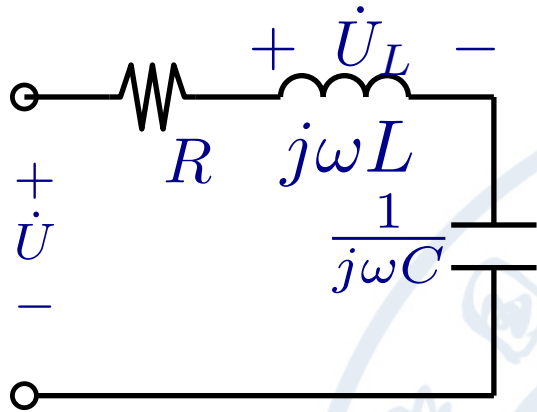
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RLC 串联电路的频率特性



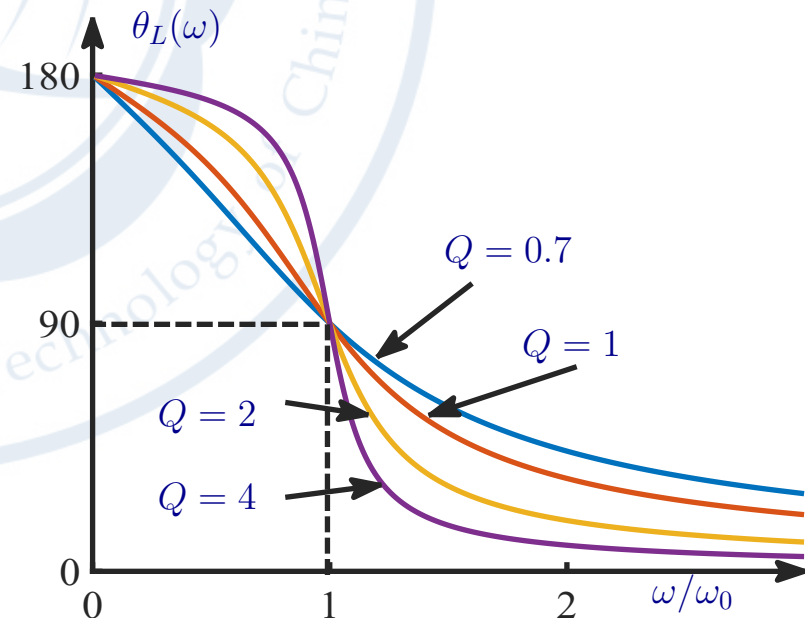
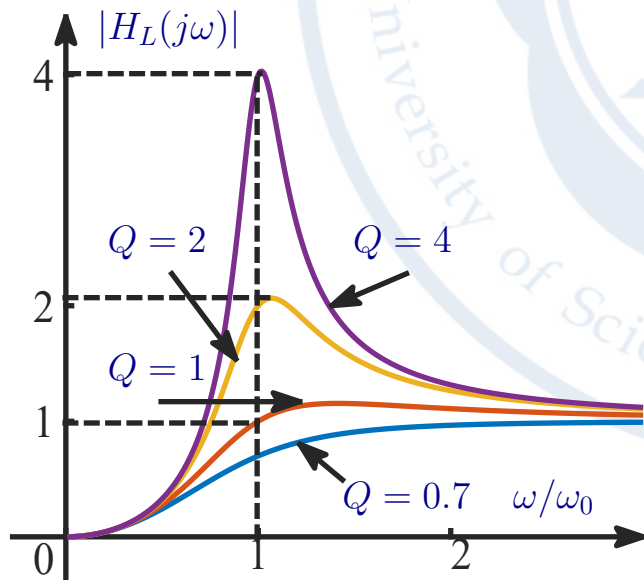
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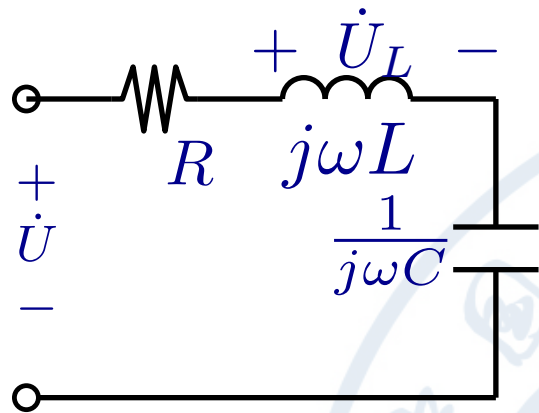
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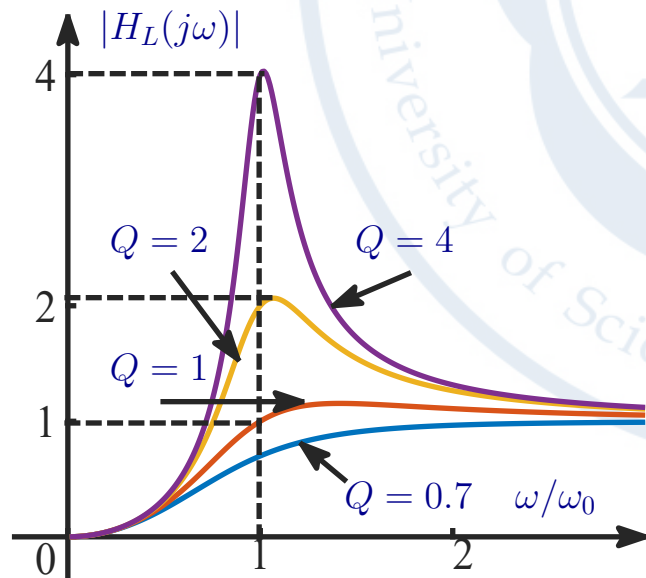
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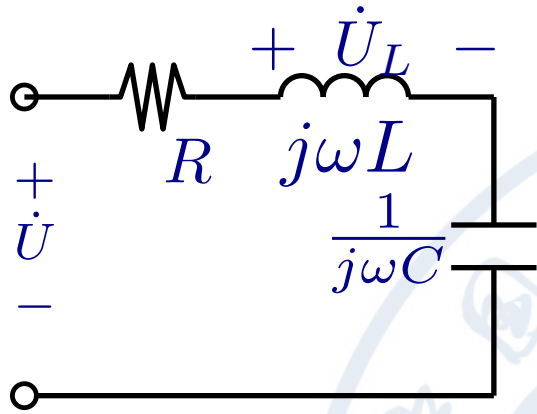
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■ 高通特性

■ 随着 Q 值越大，最大值越来越大，位置越来越接近于 ω_0

RLC 串联电路的频率特性

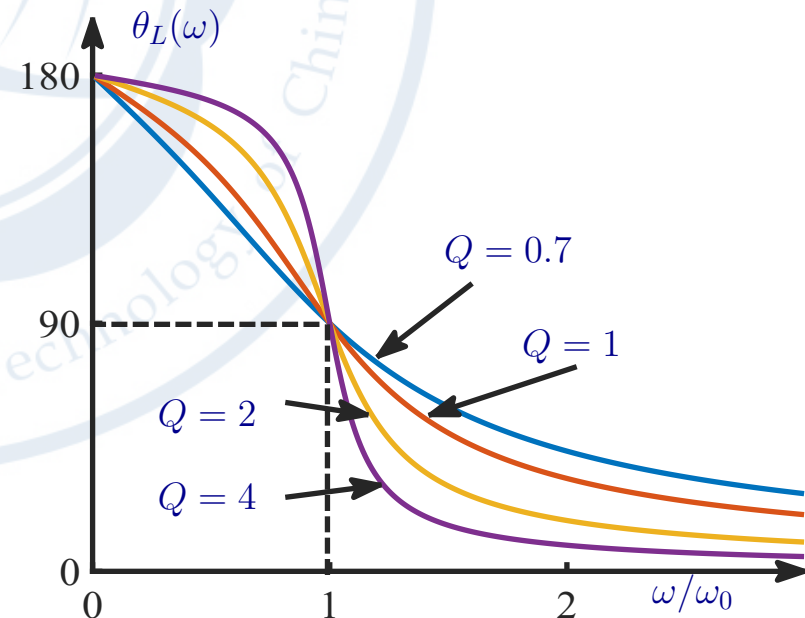


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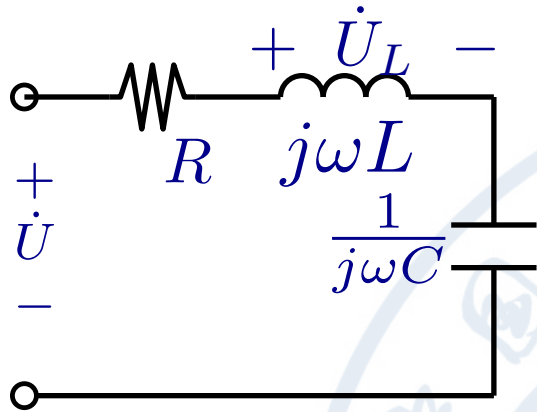
相位频率响应:

$$\theta_C(\omega) = -\arctan \frac{1}{Q\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)}$$

- $\omega/\omega_0 \rightarrow 0, \theta_L(\omega) = 0^\circ$
- $\omega/\omega_0 \rightarrow 1, \theta_L(\omega) = 90^\circ$
- $\omega/\omega_0 \rightarrow \infty, \theta_L(\omega) = 180^\circ$



RLC 串联电路的频率特性



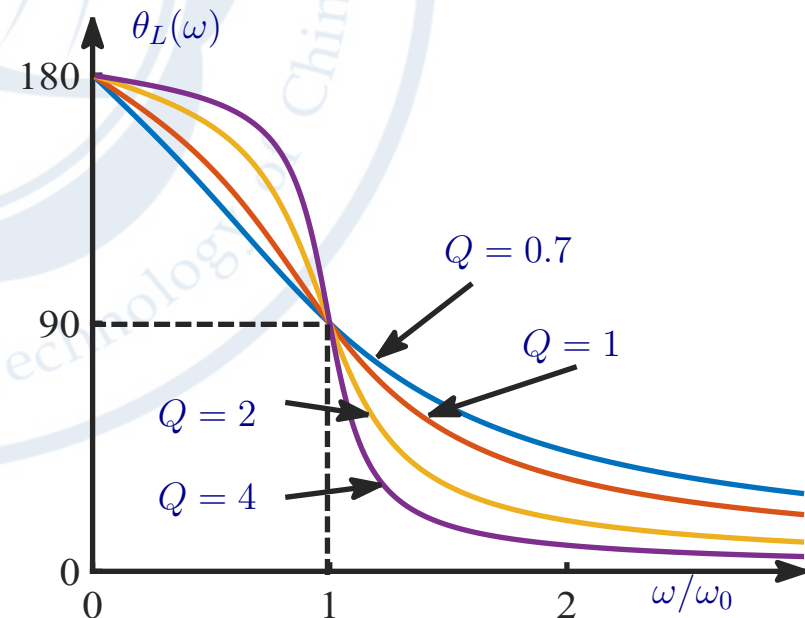
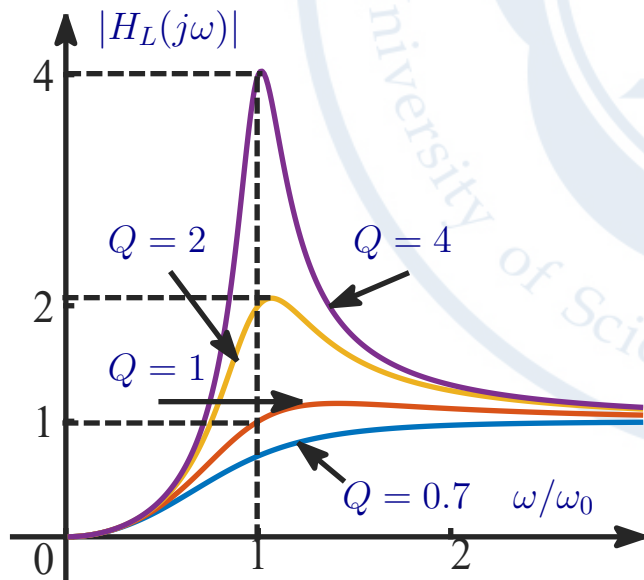
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RLC 串联电路的频率响应举例

设计 RLC 带通滤波器。已知 $R = 20\Omega$ ，要求谐振频率 $f_0 = 10^4\text{Hz}$ ，带宽 $\Delta f = 10^3\text{Hz}$ 。求电感 L 和电容 C 的数值。以及低频截止频率和高频截止频率。

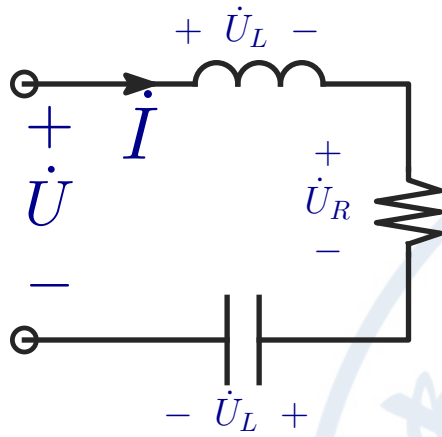
$$[1] Q = \frac{\omega_0}{\Delta\omega} = 10$$

$$[2] Q = \omega_0 L / R \Rightarrow L = \frac{QR}{\omega_0} = 3.18\text{mH}$$

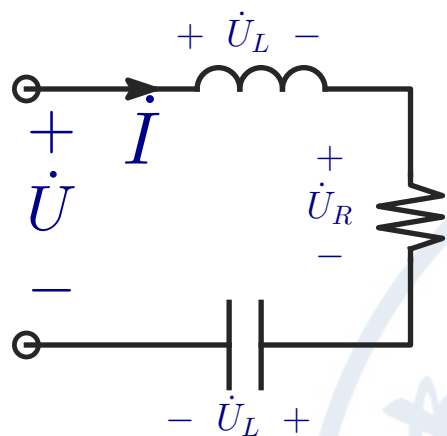
$$[3] Q = \frac{1}{\omega_0 RC} \Rightarrow C = \frac{1}{QR\omega_0} = 7.9 \times 10^{-8}\text{F}$$

$$[4] f_{c1,c2} = f_0 \left(\pm \frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right) = (1.0 \pm 0.05) \times 10^4\text{Hz}$$

串联谐振电路

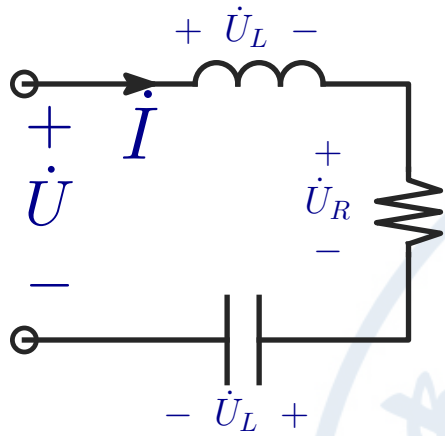


串联谐振电路



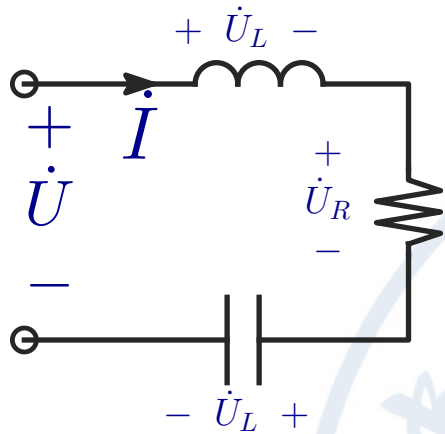
- 谐振：任何含有储能元件的一端口电路，在一定条件下可以呈现**纯电阻特性**，即端口电压和电流**同相位**。

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串联谐振电路

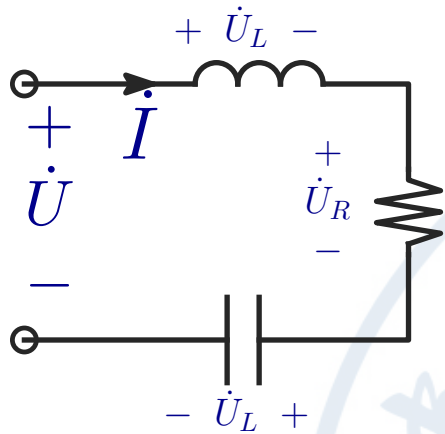


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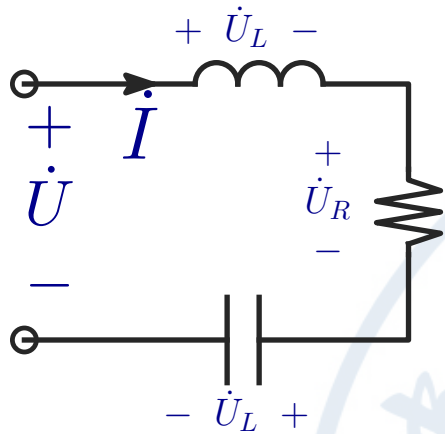
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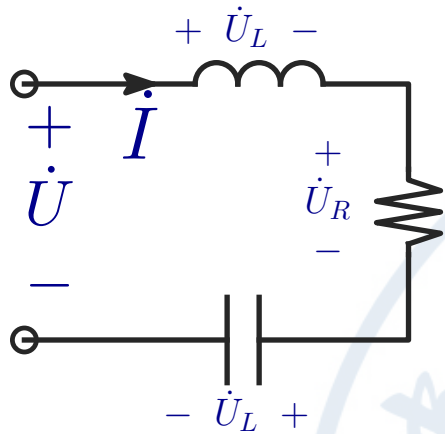
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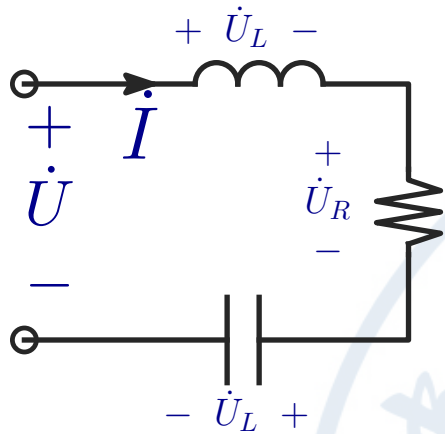
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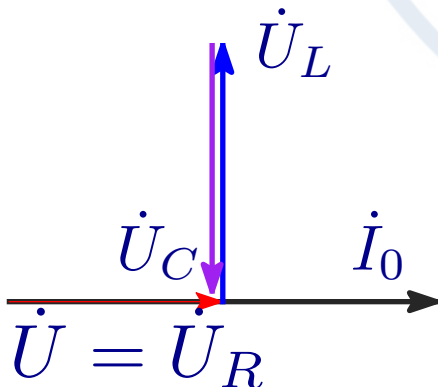
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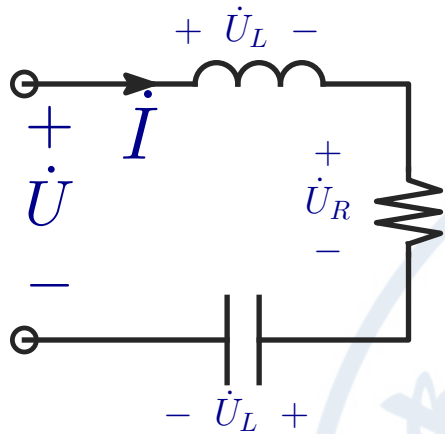


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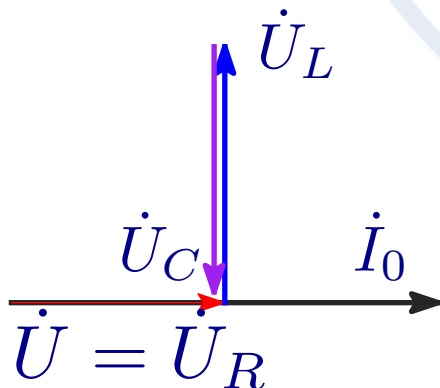


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$R \rightarrow 0, Q \rightarrow \infty$ ，谐振时类似于短路。
这称为**理想 LC 串联谐振电路**

串联谐振电路举例

一个线圈与电容串联，线圈电阻 $R = 16.2\Omega$, $L = 0.26\text{mH}$ 当 $C = 100\text{pF}$ 串联谐振。1. 求谐振频率和品质因数；2. 外加电压为 $10\mu\text{V}$ 时，频率等于谐振频率求电路中的电流和电容电压；3. 外部电压为 $10\mu\text{V}$ ， $\omega = 1.1\omega_0$ 时，求电容电压。



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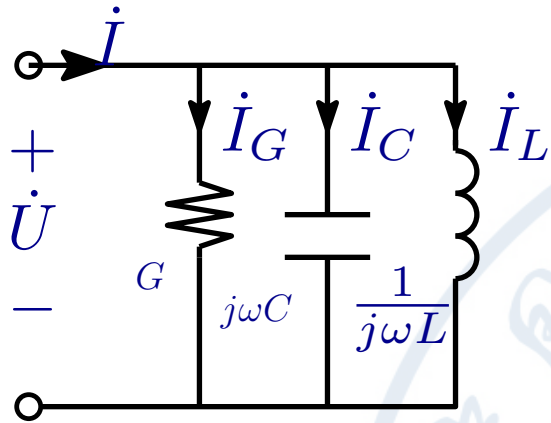
$$2. I = \frac{U_R}{I_R} = \frac{U}{R} = 0.617\mu\text{A}$$

$$3. (X'_L + X'_C)|_{\omega=1.1\omega_0} = \omega L - \frac{1}{\omega C} = 1.1\omega_0 L - \frac{1}{1.1\omega_0 C}$$
$$\frac{1}{\omega_0 C} = \omega_0 L = QR = 99.5 \times 16.2\Omega$$

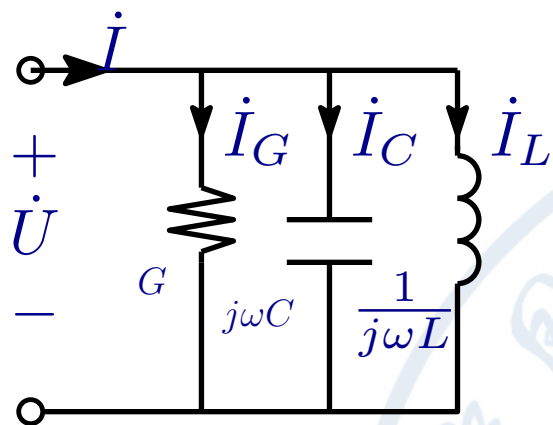
$$X'_L + X'_C = 307\Omega \rightarrow |Z| = \sqrt{R^2 + (X'_L + X'_C)^2} = 320\Omega$$

$$U_C = |X'_C| \times \frac{U}{|Z|} = 0.046\text{mV}$$

并联谐振电路

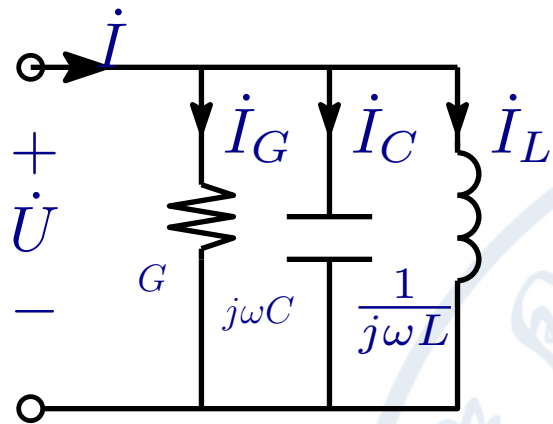


并联谐振电路



$$\dot{I} = \dot{U} \left(G + j\omega C + \frac{1}{j\omega L} \right) = \dot{U} \left[G + j \left(\omega C - \frac{1}{\omega L} \right) \right]$$

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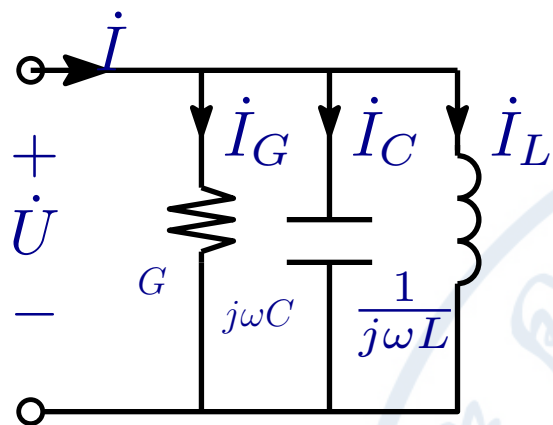


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谐振产生条件:

$$\phi_u = \phi_i \rightarrow \omega_0 C = \frac{1}{\omega_0 L} \rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

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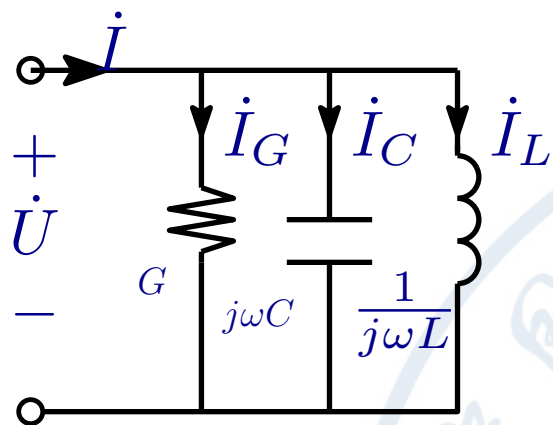
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给定电流时, 谐振时电压达到最大:

$$\dot{U} = \frac{\dot{I}}{Y} \rightarrow U = \frac{I}{Y} \leq \frac{I}{G}, \text{ when } \omega = \frac{1}{\sqrt{LC}} \text{ holds}$$

并联谐振电路



$$\dot{I} = \dot{U} \left(G + j\omega C + \frac{1}{j\omega L} \right) = \dot{U} \left[G + j \left(\omega C - \frac{1}{\omega L} \right) \right]$$

谐振产生条件:

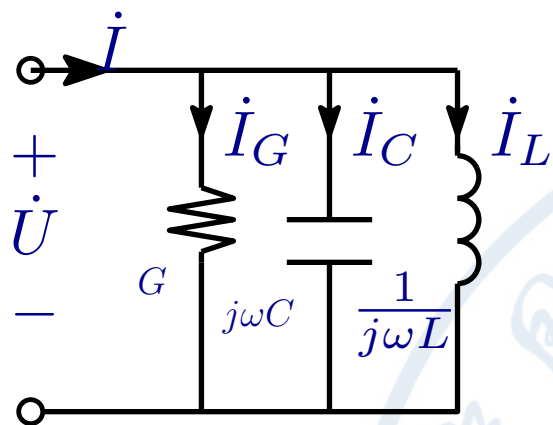
$$\phi_u = \phi_i \rightarrow \omega_0 C = \frac{1}{\omega_0 L} \rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

给定电流时, 谐振时电压达到最大:

$$\dot{U} = \frac{\dot{I}}{Y} \rightarrow U = \frac{I}{Y} \leq \frac{I}{G}, \text{ when } \omega = \frac{1}{\sqrt{LC}} \text{ holds}$$

■ 谐振时, 电容电流为 $I\omega_0 C/G$, 电感电流为 $I/(G\omega_0 L)$

并联谐振电路



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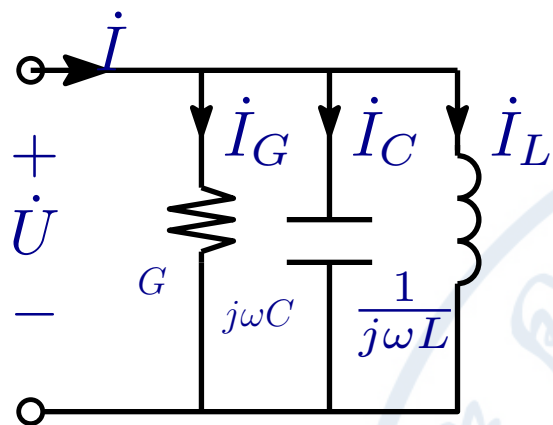
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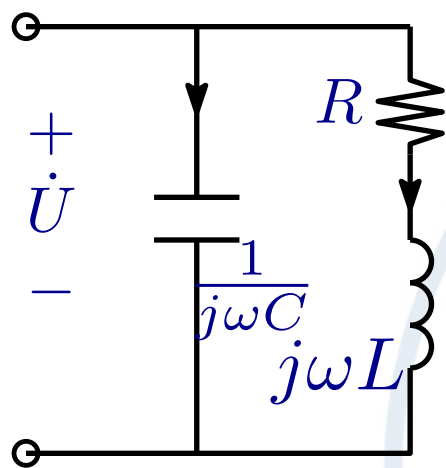
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- 品质因数 $Q = I_L/I = I_C/I = \omega_0 C/G$

并联谐振电路

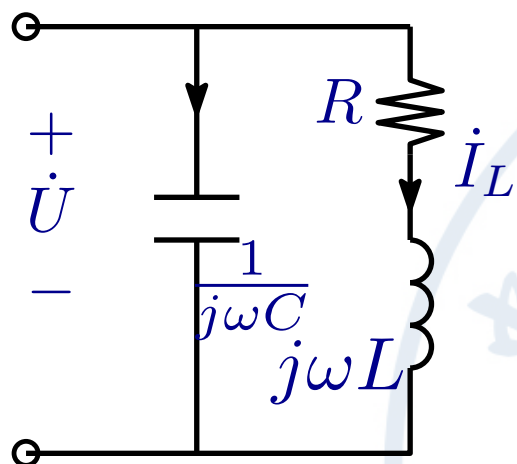
电感线圈和电容器可以并联形成谐振电路，判据为 $\phi_u = \phi_i$.



$$Y = \frac{1}{R + j\omega L} + j\omega C = \frac{R}{R^2 + (\omega L)^2} + j\left(\omega C - \frac{\omega L}{R^2 + (\omega L)^2}\right)$$

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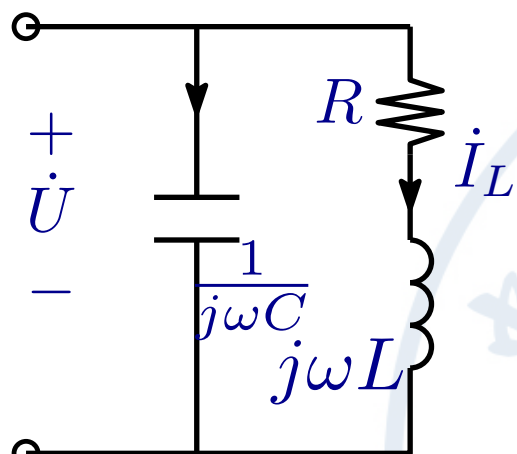
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$$C = \frac{L}{R^2+(\omega L)^2}$$

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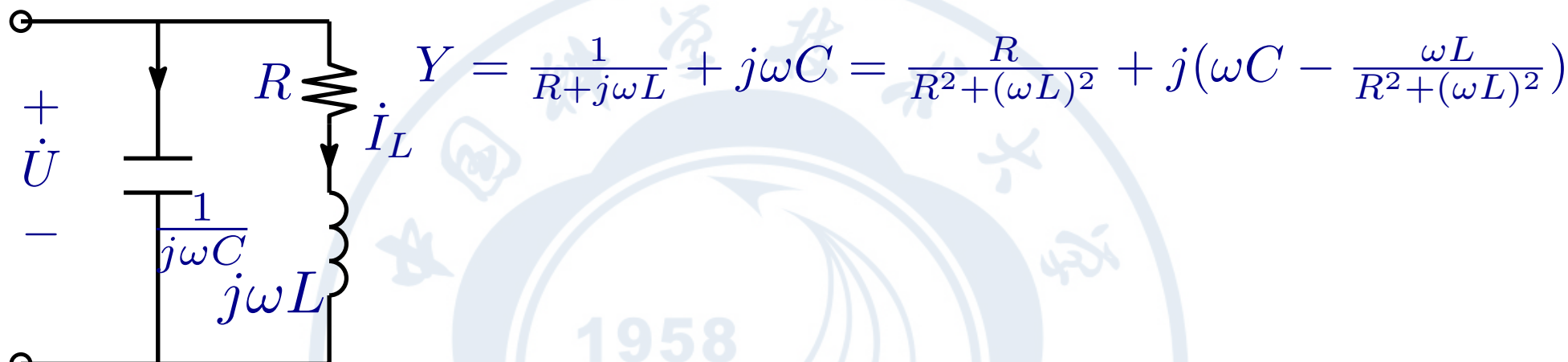
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$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

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■ 谐振时的等效电阻（R 越大，等效电阻越小）：

$$R_0 = \frac{R^2 + (\omega_0 L)^2}{R} = \frac{L}{RC}$$